

Exp No : 3a

```
Command Window

Select the incidence of element 5 with node 3:

Select the incidence of element 5 with node 4:

Enter the reference node number: 4

The reduced bus incidence matrix is:

1 -1 0
1 0 -1
1 0 0
0 1 0
0 0 1

fx >>
```

Exp No : 3b

```
Command Window

Is element 5 incident with loop 2?
Enter 1 for YES and 0 for NO: 1
Is the element 5 directed **with** the loop 2?
Enter 1 for YES and 0 for NO: 1

The Loop Incidence Matrix is:

1 0
0 1
-1 -1
1 0
0 1

fx >>
```

Exp no : 4a

```
Command Window

Is node 6 connected to branch 6?
Enter 1 for YES, 0 for NO: 1
Is the direction of branch 6 **towards** node 6?
Enter 1 for YES, 0 for NO: 0

The Branch-Path Incidence Matrix is:

-1 1 0 0 0 0
0 -1 1 0 0 0
0 0 -1 1 0 0
0 0 0 0 1 -1
0 0 0 1 -1 0
0 1 0 0 0 -1

fx >>
```

Exp No : 4b

```
Command Window
Is element 6 directed **along** the direction of cutset 4?
Enter 1 for YES, 0 for NO: 0

Is element 6 present in cutset 5?
Enter 1 for YES, 0 for NO: 0

The Cutset Matrix is:
 1  0  0  0  0
 0  1  0  0  0
 0  0  1  0  0
 0  0  0  1  0
 0  0  0  0  1
 0  1  0 -1  0

fx >>
```

Exp No 5

```
Command Window
>> expno5
Gauss-Seidel converged in 12 iterations

Bus Voltages (magnitude and angle):
Bus 1: |V| = 1.0500, Angle = 0.0000 degrees
Bus 2: |V| = 0.9818, Angle = -3.5035 degrees
Bus 3: |V| = 1.0012, Angle = -2.8624 degrees

ans =

    1.0500 + 0.0000i
    0.9800 - 0.0600i
    1.0000 - 0.0500i

fx
```

Exp No 6

```
Command Window
Converged in 10 iterations
Bus 1: V = 1.0100 p.u., Angle = 0.0000 deg
Bus 2: V = 1.0300 p.u., Angle = 1525.8812 deg
Bus 3: V = 0.0924 p.u., Angle = 1574.3123 deg

fx >>
```

Exp No 7

```
Command Window

Bus Voltages and Angles:

   Bus   Voltage_pu   Angle_deg
   ---   -
   1      1.06         0
   2      1.0097      -0.11415
   3      1.04         1.2566

fx >> |
```

Exp No 9

```
Command Window

Economic Dispatch Results:
Generator 1: 423.68 MW
Generator 2: 265.79 MW
Generator 3: 160.53 MW
Total Generation: 850.00 MW
Total Cost: 7112.24

fx >>
```

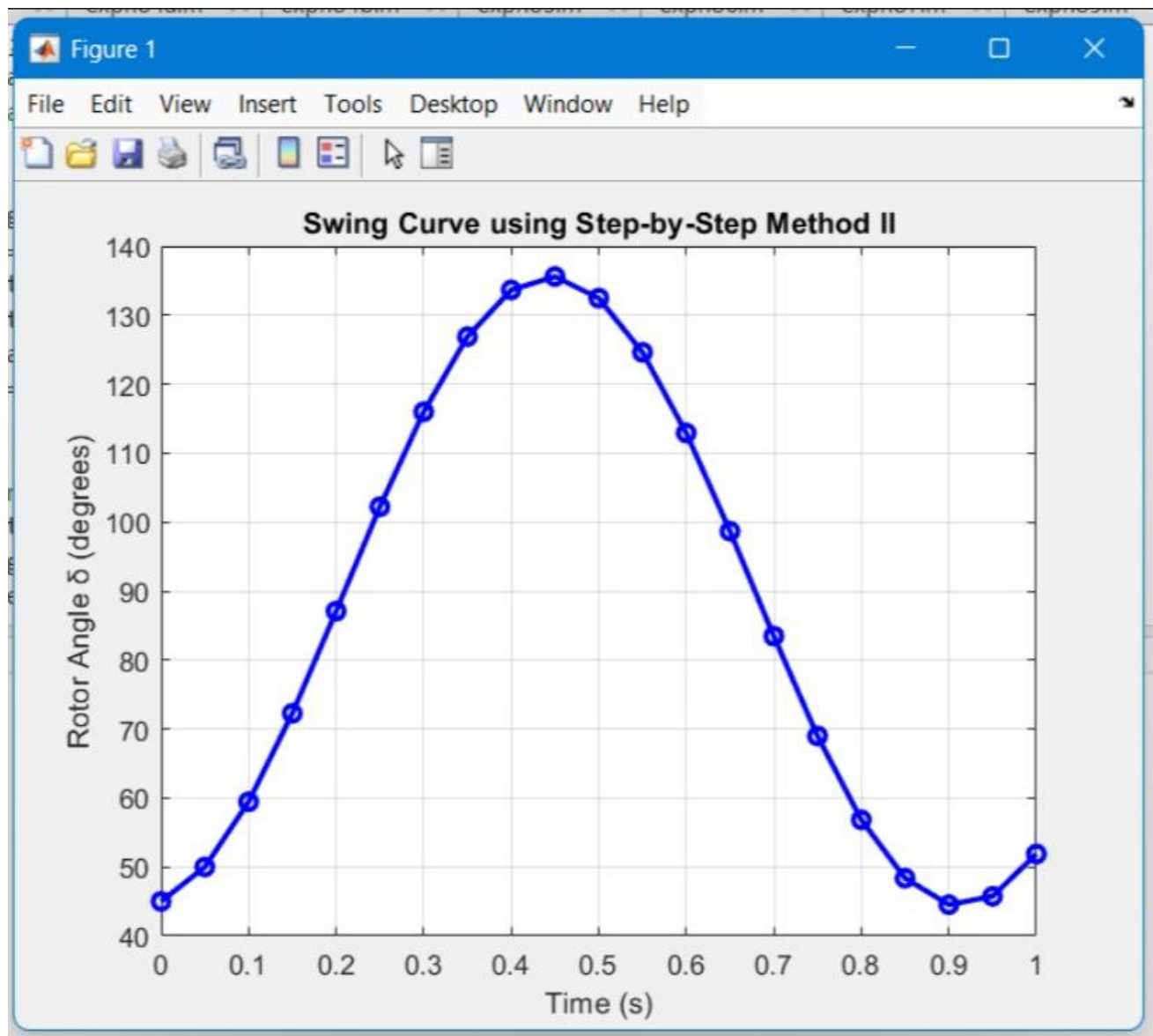
Exp No 13a

```
Command Window

Select fault type:
1. Symmetrical Fault
2. Unsymmetrical Fault
1
Symmetrical Fault
Enter the reactance for generator (pu): 0.2
Enter the reactance for transformer (pu): 0.1
Enter the reactance for motor (pu): 0.3
Enter the current value (pu): 1.0
Enter the terminal voltage of generator (pu): 1.0
Enter the power factor angle (degrees): 30
Fault current (pu): 7.8497∠4.87°

fx >>
```

Exp No 10



Exp No 13b – 1

```
Command Window

Select fault type:
1. Symmetrical Fault
2. Unsymmetrical Fault
2
Unsymmetrical Fault

Select fault type:
1. Single Line-to-Ground Fault
2. Line-to-Line Fault
3. Double Line-to-Ground Fault
1
Enter Z0 (pu): 0.1
Enter Z1 (pu): 0.2
Enter Z2 (pu): 0.2
Single Line-to-Ground Fault
Fault current = 5020.44 A
Line voltages (kV): Vab = 7.31, Vbc = 13.81, Vca = 7.31
fx >>
```

Exp no 13b – 2

```
Command Window

Select fault type:
1. Symmetrical Fault
2. Unsymmetrical Fault
2
Unsymmetrical Fault

Select fault type:
1. Single Line-to-Ground Fault
2. Line-to-Line Fault
3. Double Line-to-Ground Fault
2
Enter Z0 (pu): 0.1
Enter Z1 (pu): 0.2
Enter Z2 (pu): 0.2
Line-to-Line Fault
Fault current = 3623.19 A
Line voltages (kV): Vab = 11.95, Vbc = 0.00, Vca = 11.95
fx >> |
```

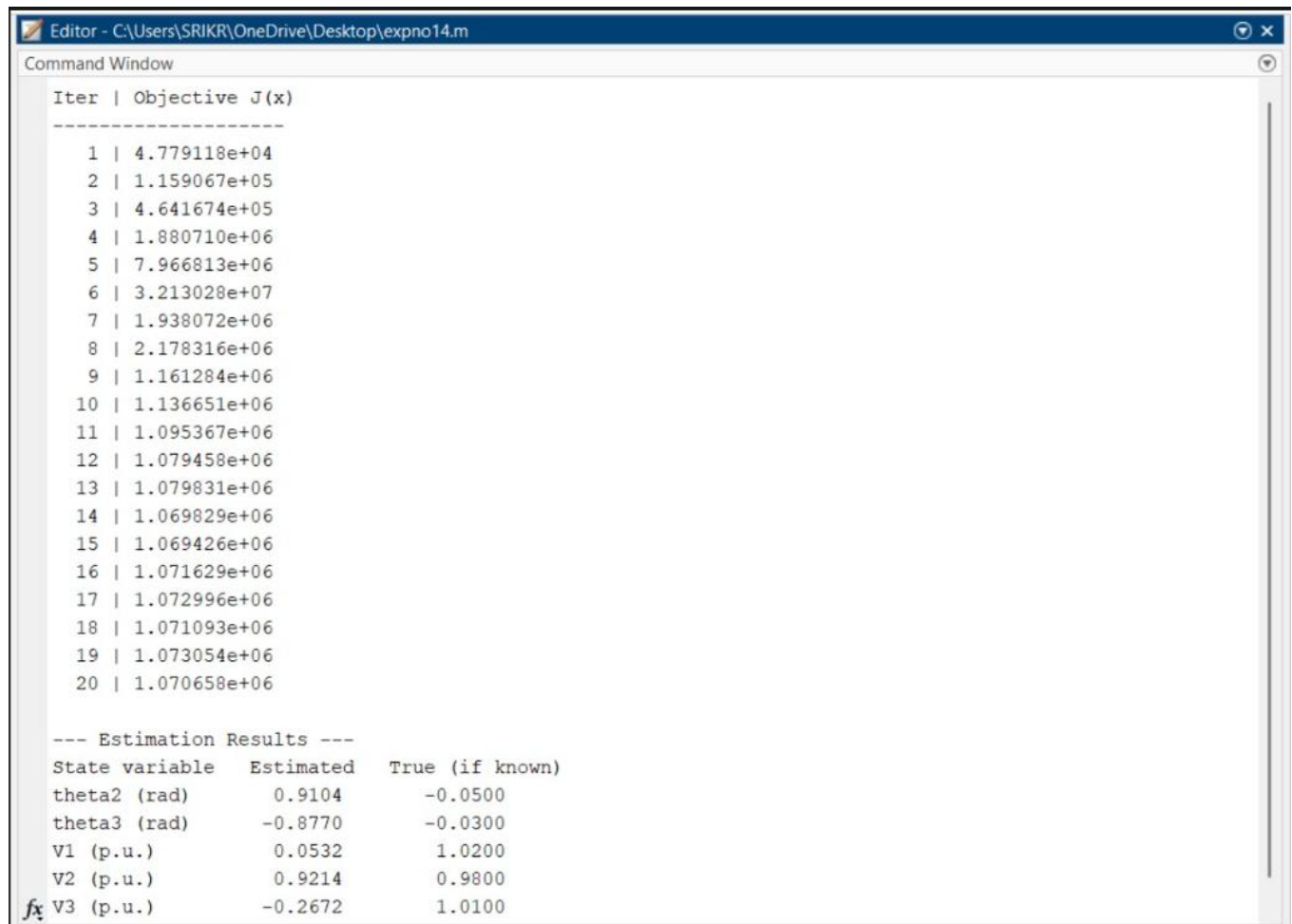
Exp no 13b – 3

```
Command Window

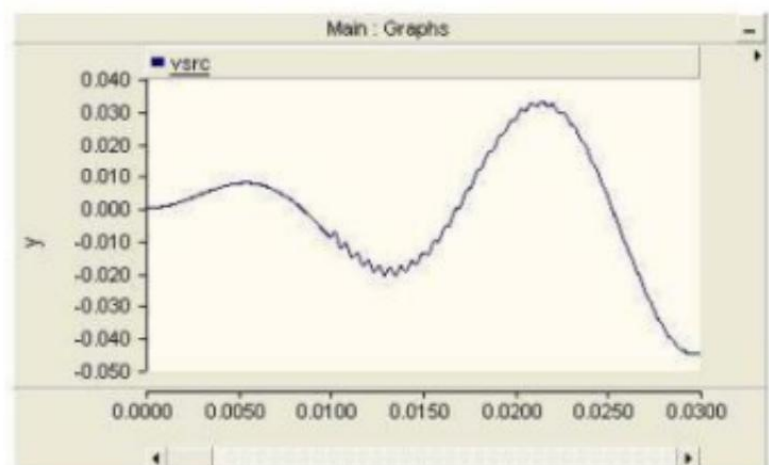
Select fault type:
1. Symmetrical Fault
2. Unsymmetrical Fault
2
Unsymmetrical Fault

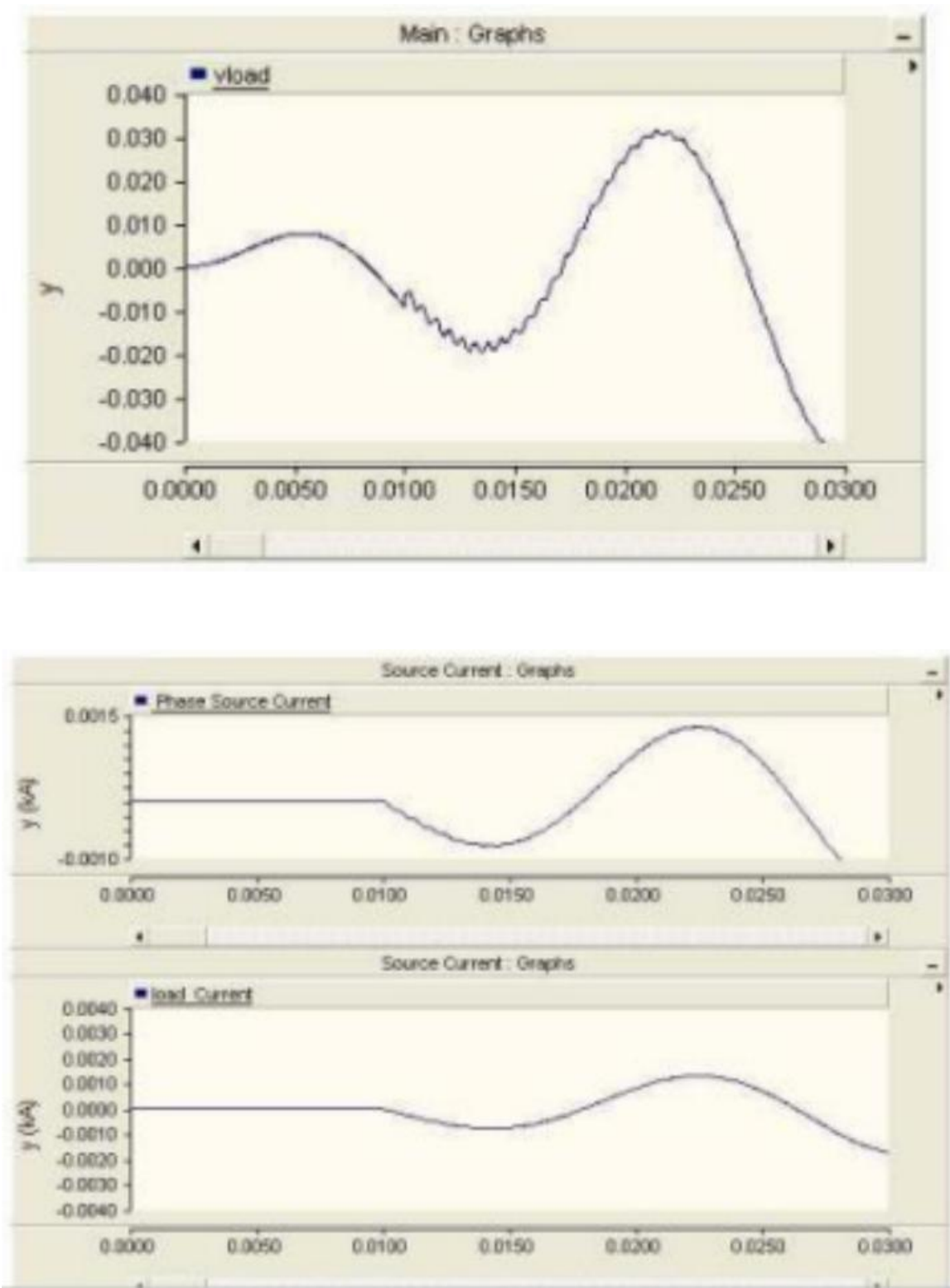
Select fault type:
1. Single Line-to-Ground Fault
2. Line-to-Line Fault
3. Double Line-to-Ground Fault
3
Enter Z0 (pu): 0.1
Enter Z1 (pu): 0.2
Enter Z2 (pu): 0.2
Double Line-to-Ground Fault
Fault current = 5434.78 A
Line voltages (kV): Vab = 5.98, Vbc = 0.00, Vca = 5.98
fx >> |
```

Exp no 14

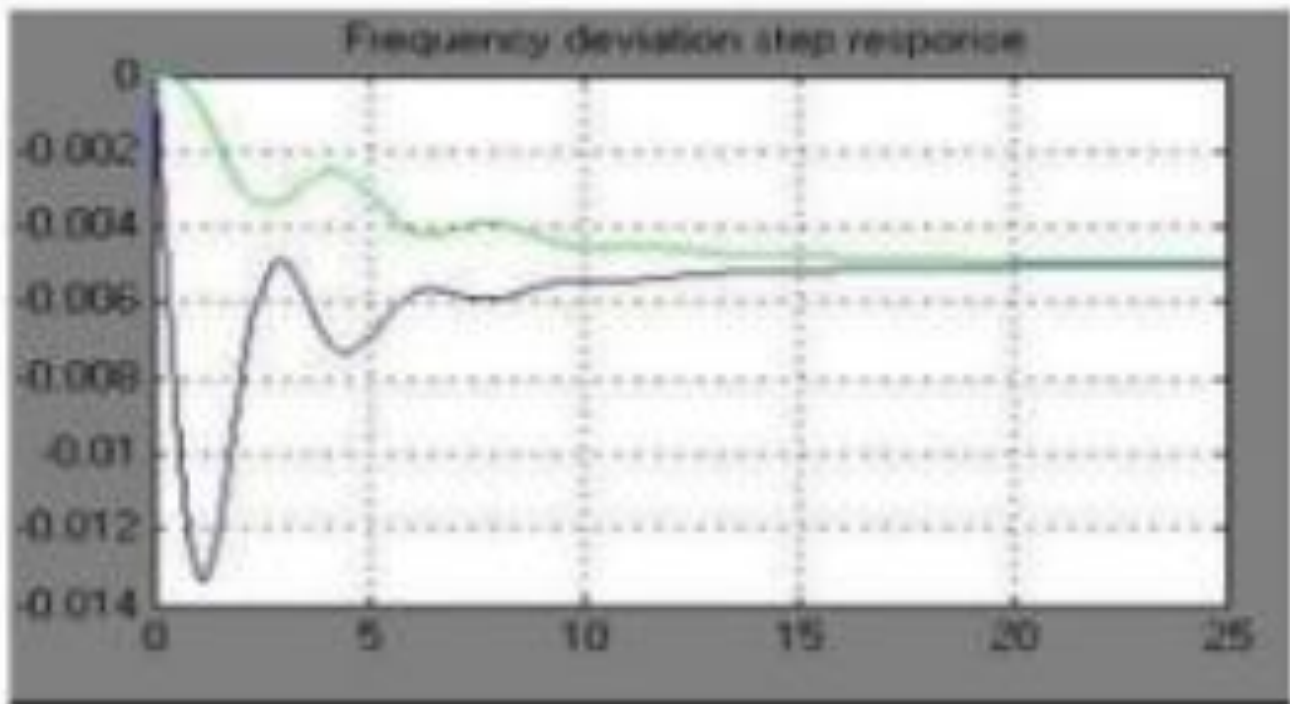
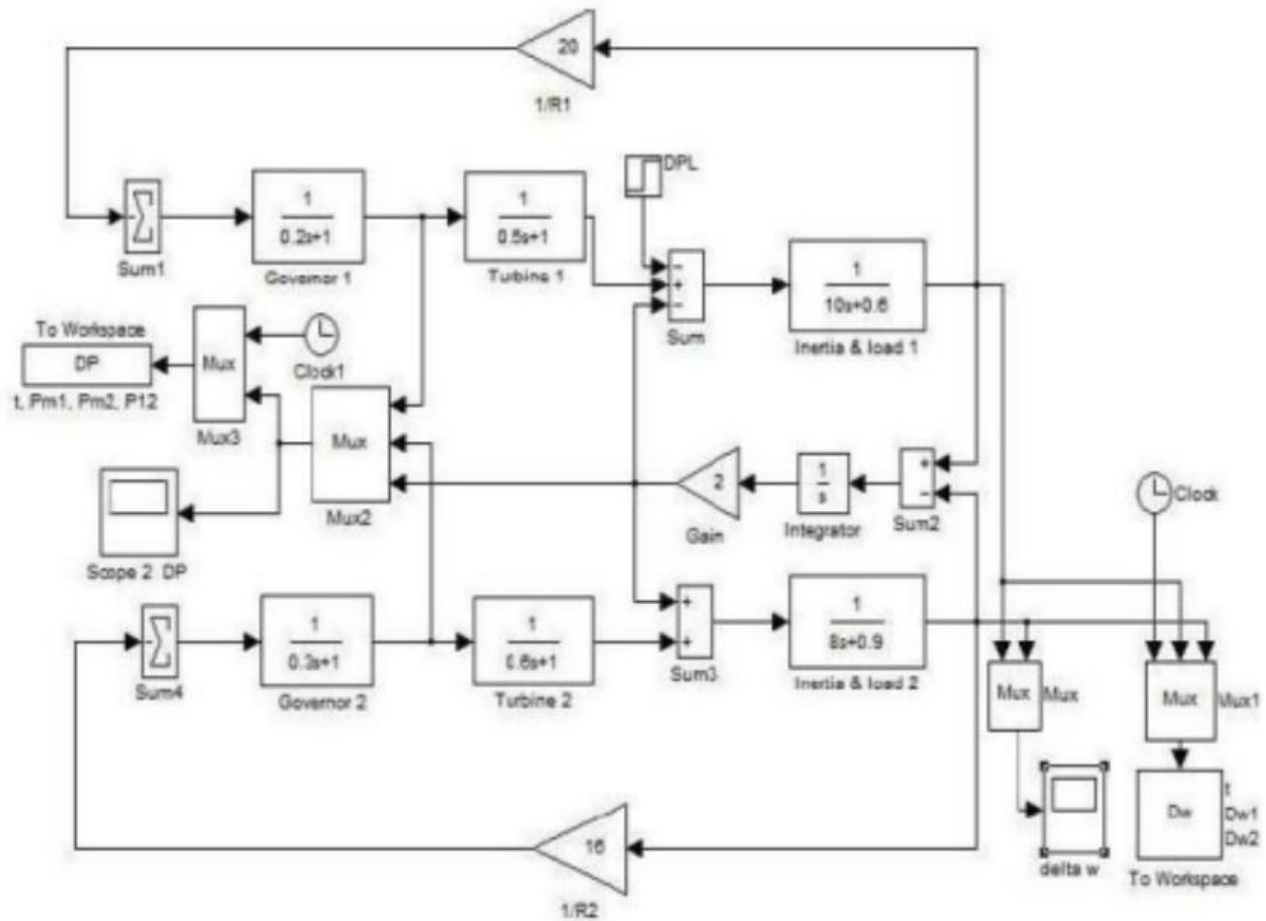


Exp no 15



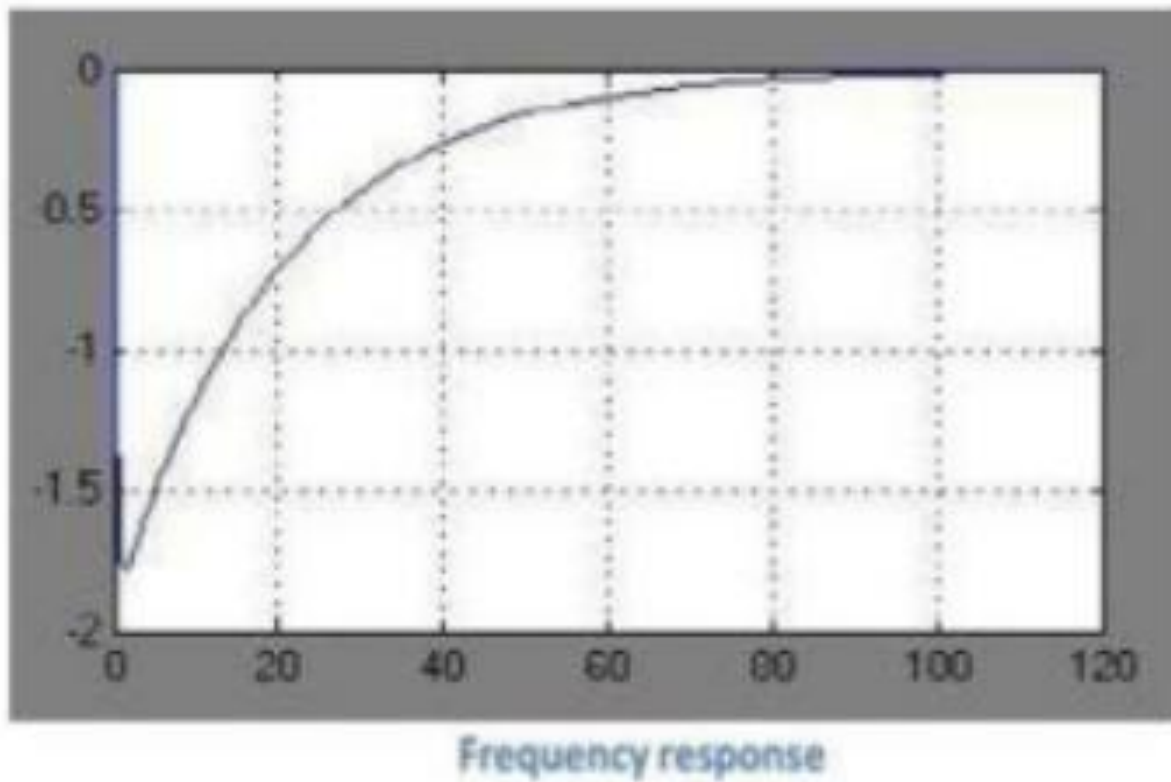
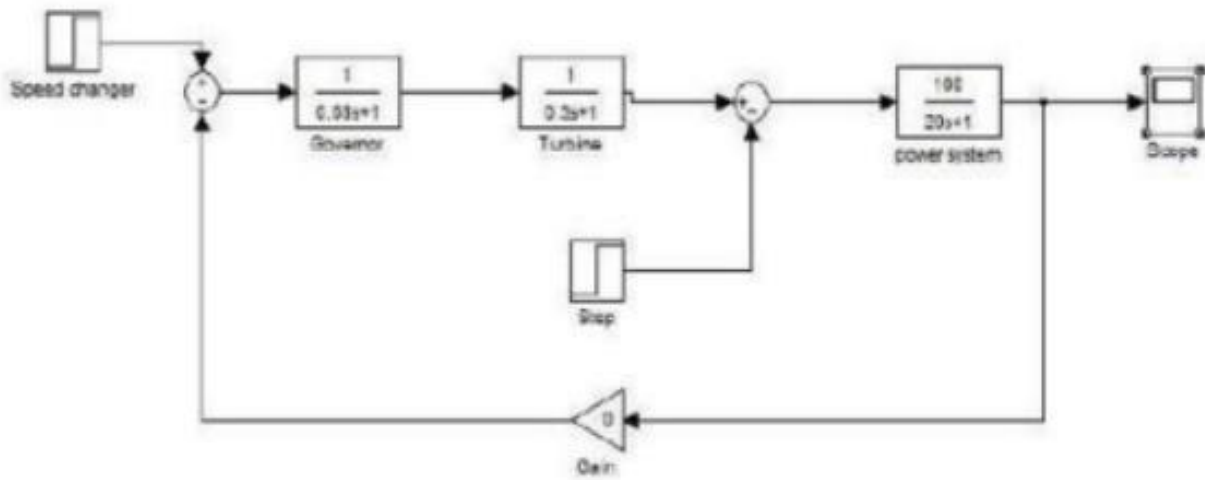


Exp no 12



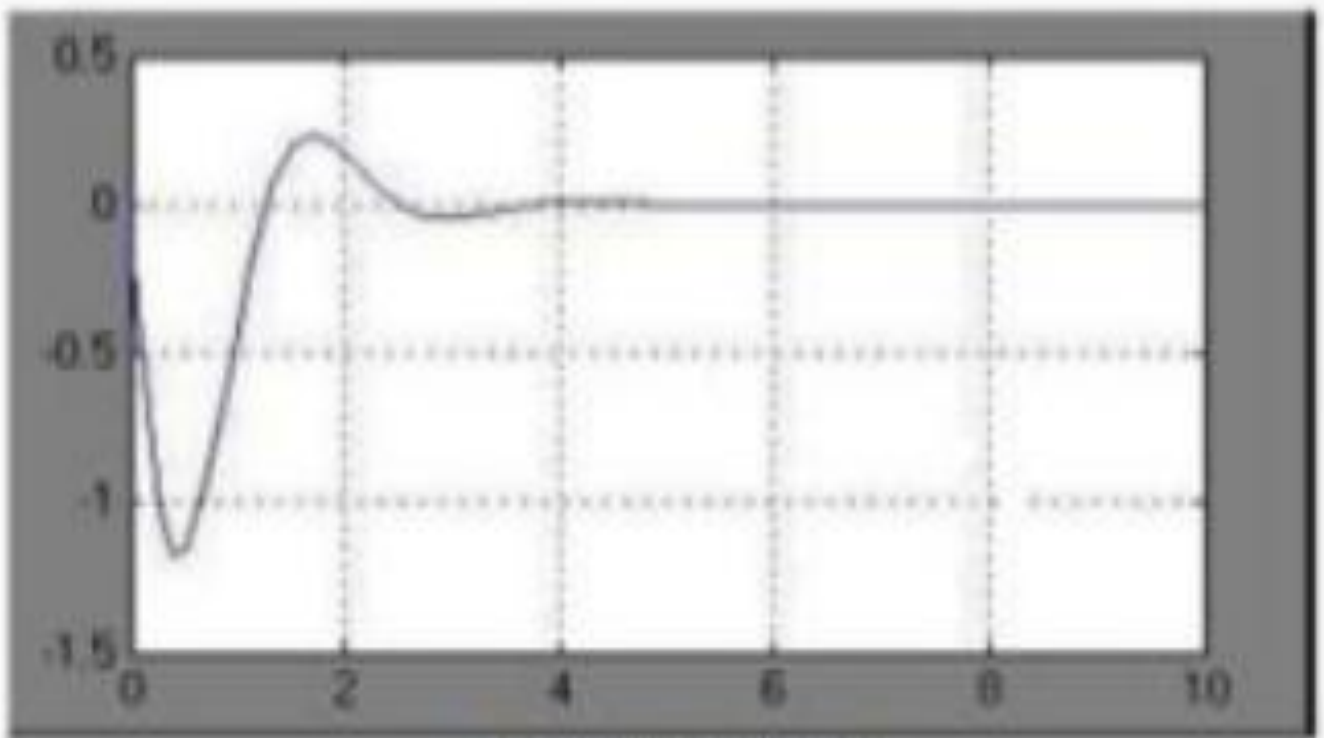
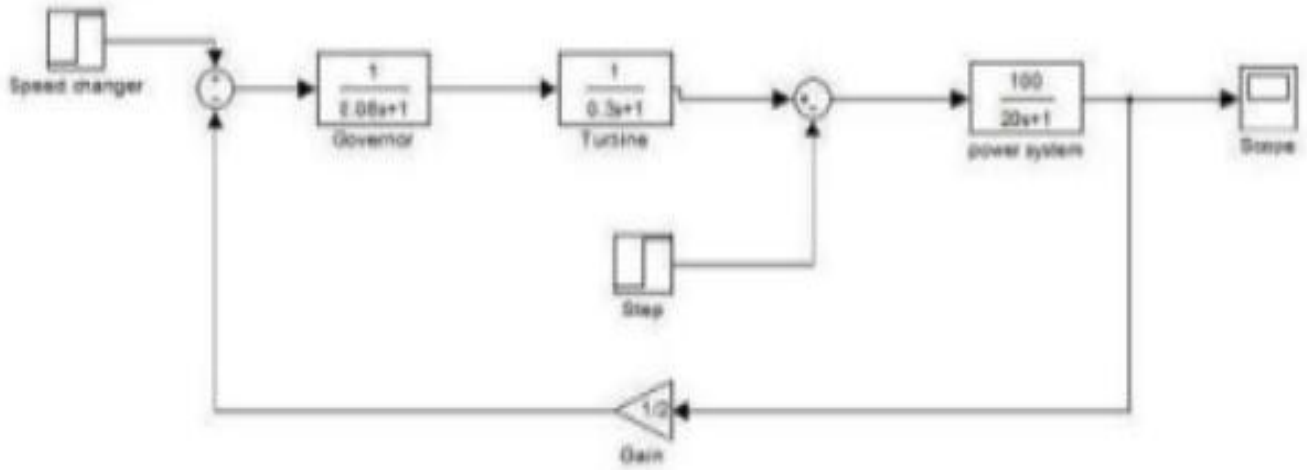
Exp No 11

WITHOUT GAIN:



Exp no 11

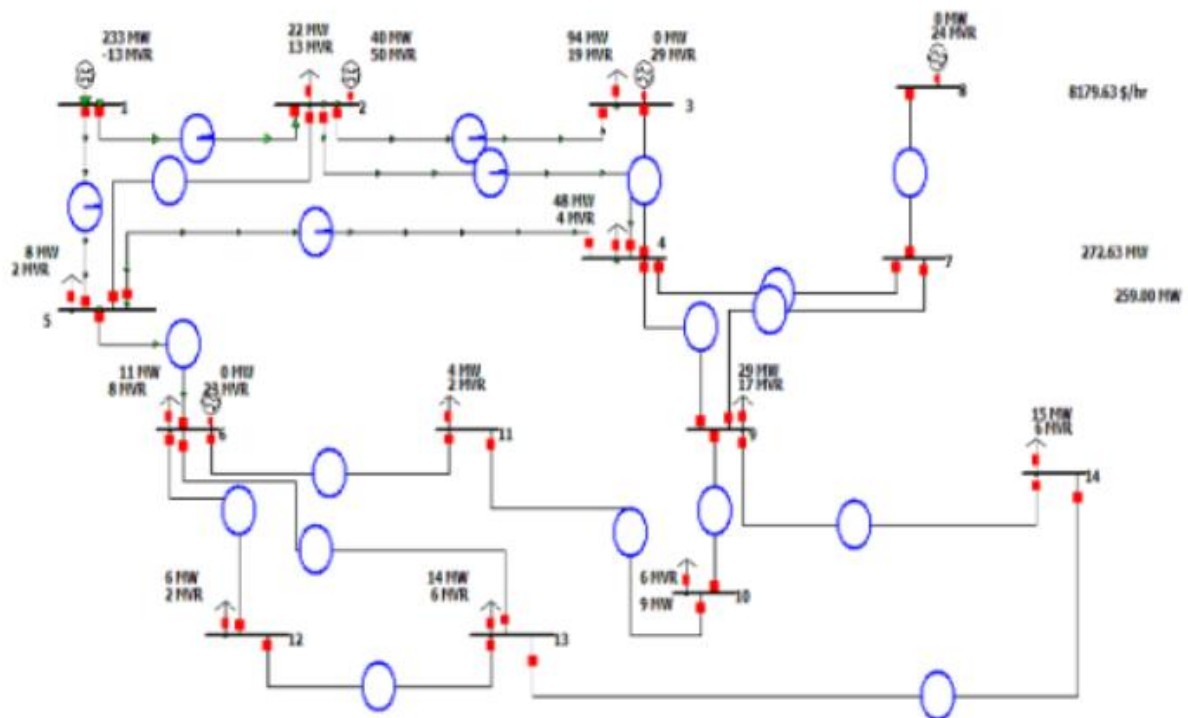
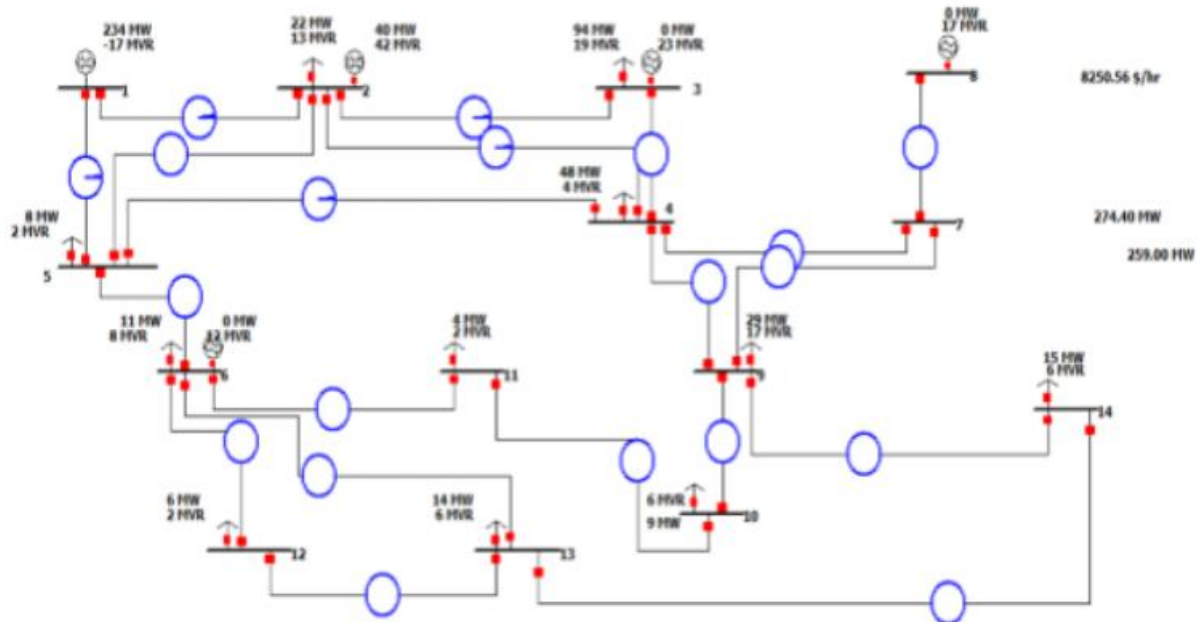
With Gain :



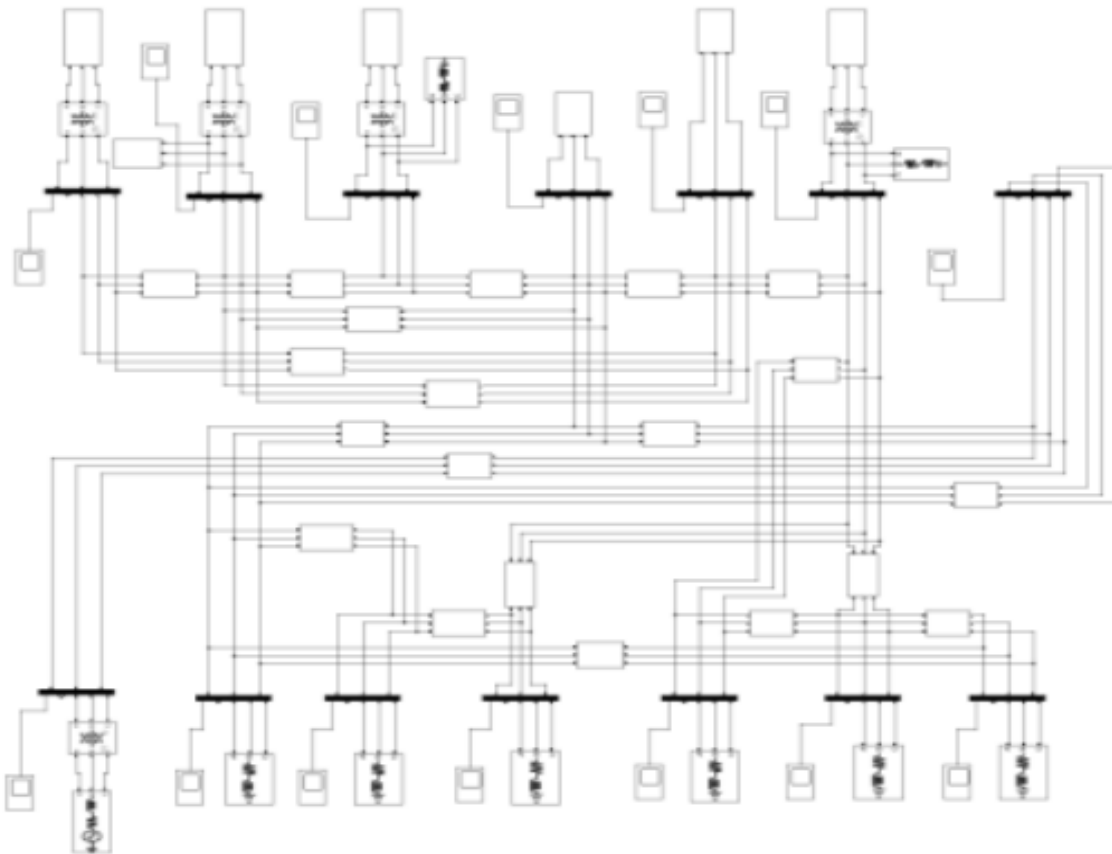
frequency response

Exp No 8 :

14 bus system :



Exp No 8



Exp No 1:

```
Command Window
Ybus =
1.3429 - 4.9809i -0.5882 + 2.3529i -0.5882 + 2.3529i 0.0000 + 0.0000i
-0.5882 + 2.3529i 3.4194 - 6.8602i -2.3922 + 1.5666i -0.7547 + 2.9412i
0.0000 + 0.0000i -0.3922 + 1.5866i 0.9980 - 3.1960i -0.5378 + 1.6420i
-0.7547 + 2.6415i -2.4390 + 0.9512i -0.5378 + 1.6420i 3.7313 - 6.7070i

fx >>
```

Exp No 2 :

```
Command Window
Zbus =
3.0642 3.3139 3.1918 3.3007 3.0298
3.3139 3.4294 3.1834 3.1246 3.1355
3.1918 3.1834 3.0452 3.8645 2.9106
3.3007 3.1238 3.3645 3.1875 2.8860
3.0948 3.1355 2.7406 2.6530 4.0096

fx >>
```